ESWCap combined C+R front end (STM32G474 carrier)

REV C: USB-only; 4-wire C/R; OPAMP3 SH buffer; J5 FH/SH/SL/FL; PA0=DRIVE, PA8=BIAS_MON

AQY221N2VY (SSOP-4) for U5–U7/U9/U10; C7=10pF C0G; R9=100R precharge; R-mode: firmware subtracts I_div=V_SH/11M (R38/R39 0.1%)

C: DRIVE -> C mux -> Rrange -> Vcap -> C1/C2 -> K1B -> N_DUT -> DUT -> B -> K1A -> GND

REV B: C1 = 1000 uF / 35 V BIPOLAR (+ marking not used).

D6 (1N4148) blocks Q1 body diode; R9 = 47R 2W; C3 returns to VREF.

Console J7: PA9=TX, PA10=RX, GND, +3V3. See docs/pcb-design.md.

R: +3V3 -> Q1 -> N_DUT -> DUT -> B -> K2A -> R mux -> RSENSE -> GND. OPAMP1: VINP=B(PA7), VINM=VREF(PA3), VOUT=PA2. TIA: LMC6482 A.

J2
1 GND
2 +3V3
3
4
5
6 DRIVE
7 VCAP_ADC
8 OP1_OUT
9 VREF
10 BIAS_SET
11 VBIAS
12 TIA_OUT
13 FL
14 C_MODE_SW
15 VN_SENSE
16
17 MUXR_EN
18
19 K2_DRV
20 SSR13
WeAct P2

J4
1 GND
2 K1_DRV
3 MUXR_A1
4 MUXR_A0
5 MUXC_EN
6 MUXC_A1
7 MUXC_A0
8 BIAS_UART_RX
9 BIAS_UART_TX
10 BIAS_EN
11
12
13 UART_RX
14 UART_TX
15 BIAS_MON
16
17
18 R_FORCE
19 SSR2_CTRL
20 VBUF
WeAct P1

J3
1 GND
2 GND
3
4
WeAct P3 (NC)

J4
1 GND
2 GND
3 +3V3
4 +3V3
WeAct P4 3V3

U2
74LVC1G34
+3V3
DRIVE 2
4 C_MUX_COM
1
3 GND

U3
C_CH3 1 Y1
C_CH2 2 Y0
MUXC_Q 3 Z1
C_MUX_COM 4 Z
MUXC_P 5 Z0
MUXC_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXC_Q
X 14 MUXC_P
X1 13 C_CH1
X0 12 C_CH0
A 11 MUXC_A0
B 10 MUXC_A0
C 9 MUXC_A1
MAX4619

R1 C_CH2 1 2 VCAP 98.9 1%
R2 C_CH1 1 2 VCAP 993.6 1%
R3 C_CH0 1 2 VCAP 98.6k 1%
R4 C_CH3 1 2 VCAP 1.01M 1%

C1 VCAP 1 2 C_BLOCK 1000uF/35V BP
C2 VCAP 1 2 C_BLOCK 68nF

D1 BAT54S
1 GND A1
K2 2 +3V3
COM 3 VCAP

J5
1 FH
2 SH
3 SL
4 FL
DUT

J6
1 V_BIAS
2 GND
V_BIAS 0–20V

R17 +3V3 1 2 VREF 7.15k
R18 VREF 1 2 GND 1k
C5 VREF 1 2 GND 100nF
R19 +3V3 1 2 VBIAS 10k
R21 VBIAS 1 2 GND 10k
C6 VBIAS 1 2 GND 100nF
C8 +3V3 1 2 GND 100nF
C9 +3V3 1 2 GND 10uF
C10 +3V3 1 2 GND 100nF

R38 SH 1 2 SH_DIV 10M 0.1%
R39 SH_DIV 1 2 GND 1M 0.1%
C20 SH_DIV 1 2 GND 100pF
C21 SH 1 2 SH_DIV 10pF

FLG1 1 +3V3
FLG3 1 GND
FLG4 1 V_BIAS

R22 SSR13 1 2 SSR1_LED 330
R23 SSR13 1 2 SSR3_LED 330
R24 SSR2_CTRL 1 2 SSR2_LED 330

U5
SSR1_LED 1 LED+
GND 2 LED-
SW 3 SSR1_SW
SW 4 V_BIAS

U6
SSR3_LED 1 LED+
GND 2 LED-
SW 3 SSR3_SW
SW 4 VCAP

U7
SSR2_LED 1 LED+
GND 2 LED-
SW 3 SSR2_SW
SW 4 FH

R9 SSR1_SW 1 2 FH 100
R11 SSR3_SW 1 2 GND 1
R10 SSR2_SW 1 2 GND 100

R12 V_BIAS 1 2 FH 4.8M
R13 +3V3 1 2 Q1_GATE 100k
R14 OP1_OUT 1 2 Q1_GATE 1k
R15 R_FORCE 1 2 Q1_GATE 10k

Q1
+3V3
S
G Q1_GATE
D Q1_D
U9
AO3401
+3V3 LED+
U9_LED_K LED-
SW SH
SW VN_IN

R20 VN_IN 1 2 VN_SENSE 100k
C3 OP1_OUT 1 2 VREF 100pF
C7 VN_SENSE 1 2 GND 10pF

U10
+3V3 LED+
U10_LED_K LED-
SW SH_DIV
SW VN_SENSE

D2 BAV199
GND 1 A1
K2 2 +3V3
COM 3 VN_SENSE

R43 C_MODE_SW 1 2 U10_LED_K 330

R41 K1_C 1 2 U9_LED_K 330

U1A
VBIAS 3 IN+
FL 2 IN-
out 1 TIA_OUT
LMC6482

U1C
+3V3
V+ 8
V- 4
GND
LMC6482

R26 K2_DRV 1 2 K2_BASE 1k
R28 K2_BASE 1 2 GND 10k
R60 SL 1 2 SL_BUF 100k

U1B
SL_BUF 5 IN+
VBUF 6 IN-
out 7 VBUF
LMC6482

D4 K1_C 2 1 +3V3 1N4148

R25 K1_DRV 1 2 K1_BASE 1k

Q2
K1_BASE 1 B
C 3 K1_C
E 2 GND
BC847

R27 K1_BASE 1 2 GND 10k

K1
+3V3 COIL+
K1_C 12 COIL-
NC1 3 GND
COM1 4 FL
NO1 5 C_BLOCK
NC2 10 FH
COM2 9
NO2 8

R35 2 GND
1 10k BIAS_EN

R16 FL 1 2 RF_TOP 10M 0.1%
C4 FL 1 2 RF_TOP 10pF

D3 BAT54S
GND 1 A1
K2 2 +3V3
COM 3 TIA_OUT

R34 BIAS_SET BIAS_SET_F
C16 BIAS_SET_F GND 1k
R36 BIAS_SET_F BIAS_MON 10M
C17 BIAS_MON GND 100nF

Q3
K2_C
C 3
B 1 K2_BASE
E 2 GND
BC847

J8
Pin_1 1 BIAS_SET_F
Pin_2 2 BIAS_EN
Pin_3 3 BIAS_MON_F
Pin_4 4 GND
Pin_5 5 +3V3
Pin_6 6 BIAS_UART_TX
Pin_7 7 BIAS_UART_RX
Pin_8 8 GND

D5 BIAS_CTRL 2 1 +3V3 1N4148

K2
+3V3 COIL+
K2_C 12 COIL-
NC1 3
COM1 4 FL
NO1 5 MUXR_COM
NC2 10 RF_TOP
COM2 9 TIA_OUT
NO2 8
KELVIN

R5 R_CH0 1 2 GND 98.9 0.1%
R6 R_CH1 1 2 GND 993.6 0.1%
R7 R_CH2 1 2 GND 98.6k 0.1%
R8 R_CH3 1 2 GND 4.7M 0.1%

U4
R_CH3 1 Y1
R_CH2 2 Y0
MUXR_Q 3 Z1
MUXR_COM 4 Z
MUXR_P 5 Z0
MUXR_EN 6 ENABLE
7 NC
GND 8 GND
VCC 16 +3V3
Y 15 MUXR_Q
X 14 MUXR_P
X1 13 R_CH1
X0 12 R_CH0
A 11 MUXR_A0
B 10 MUXR_A0
C 9 MUXR_A1
MAX4619

C11 +3V3 1 2 GND 100nF
C12 +3V3 1 2 GND 100nF
C13 +3V3 1 2 GND 100nF
C14 +3V3 1 2 GND 100nF
C15 +3V3 2 GND 10uF

R29 DRIVE 1 2 GND 10k
R30 C_MODE 1 2 +3V3 100k
R31 VN_EN 1 2 +3V3 100k
R32 VCAP 1 2 VCAP_ADC 1k

R33 VCAP 1 2 C_BLOCK 1M
D6 Q1_D 2 1 FH 1N4148

J7
1 UART_TX
2 UART_RX
3 GND
4 +3V3
Console

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File: Project.kicad_sch

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